

SIRISH CHEJERLA

Allen, TX, USA | +1 2247161859 | sirish.chejerla.dev@gmail.com | LinkedIn

SUMMARY

Full Stack Developer and **AI Software Engineer** with 4+ years delivering production web applications and **LLM**-powered back-end systems across telecom, payroll, and enterprise software contributing to platforms serving 800+ enterprise customers and processing 30,000+ daily transactions. Full-stack depth spans **React**, **TypeScript**, **Java** (**Spring Boot**), and **Python** (**FastAPI**) across **REST API** design, **microservices architecture**, **event-driven architecture**, **PostgreSQL**, **MongoDB**, **OAuth 2.0/JWT**, **Docker**, **Kubernetes**, **AWS**, **Azure**, and **CI/CD** pipelines. AI engineering work includes shipped **Generative AI (GenAI)** systems **Retrieval-Augmented Generation (RAG)** pipelines using **LangChain**, **LangGraph**, **FAISS**, **agentic AI** orchestration, **prompt engineering**, **semantic search**, and **vector databases** (Pinecone, pgvector) cutting mean-time-to-root-cause from 4 hours to 45 minutes in a live telecom operations pipeline.

EXPERIENCE

Capgemini

Sep 2025 – Present

AI Software Engineer (Client: AT&T Plano Campus)

Plano, TX

- Built an **LLM**-powered log triage pipeline in **Python** using a self-hosted **LLM** (Llama 3.1 via **vLLM**) that classifies 150+ daily network alerts, cutting mean-time-to-root-cause from 4 hours to under 45 minutes for a 10-engineer AT&T operations team.
- To eliminate manual runbook search across 120+ documents, designed an end-to-end **Retrieval-Augmented Generation (RAG)** system using **LangChain**, **FAISS**, and **PostgreSQL** with **prompt engineering** and **semantic search** reducing time-to-remediation from 15 minutes to under 10 seconds per query.
- Containerized 6 **FastAPI** **microservices** on **AWS EKS** with **Docker** and **Kubernetes**, implementing **distributed tracing** via Prometheus and Grafana for **observability** sustaining 400 concurrent provisioning requests while meeting p99 latency SLAs across 3 AT&T network teams.
- Manual review of 200+ daily network configuration scripts consumed 3 engineer-hours per day; automated the validation pipeline using **Python** and Paramiko **SSH**, reducing that to a 20-minute scheduled check with zero manual intervention.
- Dropped average **REST API** response time from 1.6 seconds to 280 milliseconds across 10 provisioning **microservices** — achieved through composite **PostgreSQL** indexes, query-plan analysis, and rewriting 3 high-cost stored procedures.
- Shipped an automated Dev Sanity unit-test image-qualification pipeline in **Python** and **Docker** that validates uploaded debug images in under 30 minutes down from a 3-hour manual process earning VP-level approval for org-wide rollout across AT&T network teams.
- Delivered a **full-stack** network monitoring dashboard (**React** frontend, **FastAPI** backend) providing real-time health visibility across 25,000 Internet Gateway devices; **AI**-driven anomaly detection triggers automated email alerts to on-call teams identifying downed services and process faults, with device logs forwarded to a centralized log store.
- Investigated 25+ production incidents using **Python**-based log parsing and self-hosted **LLM** diagnostics, cutting average resolution time from 4 hours to 90 minutes and surfacing recurring fault patterns that expanded the **RAG** runbook knowledge base by 15 entries.

Paychex USA

Jul 2025 – Sep 2025

Full Stack Developer

USA

- Rebuilt 2 legacy batch-processing modules in **Python** with **concurrent execution** and **asynchronous processing**, raising payroll throughput from 25,000 to 45,000 transactions per hour during end-of-quarter cycles.
- Developed **React** and **TypeScript** dashboards for 2 payroll management modules, reducing HR administrator data-entry steps from 8 to 5 for workflows covering 4,000+ employee records.
- Integrated **OAuth 2.0** and **JWT** token-based authentication across 3 application modules, securing access for 4,000+ employee accounts and satisfying **SOC 2** audit requirements.
- Replaced 3 point-to-point integrations with 4 **Python** **REST API** endpoints connecting 6 internal business applications, cutting inter-service round-trip latency from 600ms to under 180ms across a **microservices** layer.
- Inconsistent local environments across an 8-engineer team caused recurring setup failures that delayed sprint starts; standardized **Docker**-based development environments and **CI/CD** pipeline config, cutting new-hire onboarding time from 2 days to half a day.

Dell Technologies

Jan 2023 – Jul 2023

Full Stack Developer

India

- Ingested and analyzed 400GB of hardware telemetry using **Python** and **PostgreSQL** **data pipelines**, surfacing 6 recurring failure patterns that informed quarterly procurement decisions across 3 data center teams.
- A 2-hour processing backlog was delaying inventory reconciliation across 2 downstream systems; refactored event-streaming consumer groups handling 80,000+ daily status updates using **event-driven architecture** patterns to clear the queue.
- To replace overnight batch reports with live visibility, created **React** and **Spring Boot** reporting dashboards for 3 global supply chain centers — giving operations managers real-time access to 15,000+ daily logistics events.
- Benchmarked **PyTorch** classification models against **C++** rule-based pipelines on 80,000 hardware records; the evaluation report guided ML tooling selection and shaped a 6-month infrastructure analytics roadmap.
- Migrated 6 legacy **Java EE** services to **Spring Cloud** **microservices**, applying **distributed systems** design principles and reducing average release cycle time from 3 weeks to 4 days.

- To close a test coverage gap across 3 **data pipeline** modules, authored 150+ **PyTest** unit and **integration tests** using **Test-Driven Development (TDD)** — reducing production defect escapes from 5 per release to 1.
- Collaborated with 2 embedded firmware engineers to validate BMC telemetry datasets, resolving 2 data format mismatches that had corrupted 3 months of historical analytics.

Genesys Cloud Services Inc.

Software Engineer

Sep 2020 – Dec 2022

Chennai, India

- Implemented **Java** and **Spring Boot** backend services for a cloud contact center platform serving 800+ enterprise customers, sustaining 30,000+ daily interactions with fewer than 4 hours of unplanned downtime per quarter via **high-availability** architecture.
- Engineered multithreaded **REST APIs** in **Java** and **Spring Boot** handling 2,000+ concurrent session requests, supporting real-time routing across 50+ enterprise **cloud-native** contact center deployments.
- Architected **MongoDB NoSQL** data models managing 300,000+ interaction records — designed for sub-second query retrieval across **high-throughput** workflows processing 40,000+ messages per month.
- Deployments across 10 **microservices** averaged 35 minutes and required manual handoffs; streamlined **Jenkins CI/CD** pipelines to cut that to 10 minutes and enable same-day hotfix releases to staging and production.
- Resolved 15+ critical **production** incidents involving thread contention and **JVM** memory leaks, reducing mean-time-to-recovery from 2 hours to under 30 minutes for cloud contact center services.
- To prepare for a compliance review, authored **Python** auditing utilities scanning 400+ security permissions across 6 services — identifying 18 misconfigured access rules and cutting audit preparation from 3 days to 1 day.
- Participated in **Agile/Scrum** sprint ceremonies across a distributed team of 8 engineers, delivering 4 major platform features over 8 sprints spanning 2 years.

PROJECTS

Real-Time Order Management System |

- Constructed a **full-stack** order management platform with a **React/TypeScript** frontend and **FastAPI** backend, supporting real-time inventory updates across 15 product categories via **WebSocket** connections.
- Wired a **Redis** caching layer between the **REST API** and **PostgreSQL**, dropping database read load from 1,200 to under 80 queries per minute during peak browsing sessions.

Intelligent Document Q&A Platform |

- Launched a **Retrieval-Augmented Generation (RAG)**-based document retrieval system using **LangChain** and **FAISS** that indexed 2,000+ technical documents, returning contextual answers in under 3 seconds per query across 12 knowledge bases.
- Indexed 150,000+ **vector embeddings** into a **FAISS** index using a self-hosted embedding model, processing 500+ document pages per batch to enable **semantic search** at sub-second retrieval speed.
- Exposed the retrieval pipeline as a **FastAPI REST API** service, handling 200+ daily queries from 3 internal teams during a 4-week pilot and cutting average document lookup time from 20 minutes to under 2 minutes.

SKILLS

Programming Languages: Python, Java (17/21), SQL, TypeScript, Go (Golang), C++, Bash/Shell Scripting

Frontend Development: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, GraphQL, Data Visualization, Responsive Web Design

Backend & System Design: FastAPI, Spring Boot, Spring Cloud, REST APIs, gRPC, GraphQL, Microservices Architecture, Distributed Systems, Event-Driven Architecture, High-Availability Systems, CQRS, Concurrency, Asynchronous Processing, WebSockets, API Design, System Design

AI/ML & Generative AI: Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, Agentic AI, Multi-Agent Systems, vLLM, Ollama, Llama 3, Prompt Engineering, Semantic Search, Vector Databases (FAISS, Pinecone, pgvector, Weaviate), Embeddings, Fine-Tuning (LoRA/QLoRA), Model Evaluation (Ragas, DeepEval), AI Observability, LangSmith, PyTorch

Cloud & DevOps: AWS (EKS, EC2, S3, Lambda, RDS, CloudWatch, Bedrock), Azure, Azure Kubernetes Service (AKS), Google Cloud Platform (GCP), Docker, Kubernetes, Terraform, Infrastructure as Code (IaC), Jenkins, GitHub Actions, CI/CD Pipelines, Prometheus, Grafana, Datadog, ELK Stack, Distributed Tracing, Observability, Serverless Architecture

Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka, RabbitMQ, Elasticsearch, MySQL, DynamoDB, pgvector, Stream Processing, Query Optimization, NoSQL, Database Design

Methodologies & Leadership: Agile, Scrum, SDLC, Test-Driven Development (TDD), System Design, Technical Leadership, Cross-Functional Collaboration, Code Review, Incident Management, Production Support, On-Call Rotation

Tools & Frameworks: Git, Linux/Unix, PyTest, JUnit, Mockito, Postman, Swagger/OpenAPI, Prometheus, Grafana, LangSmith, Jira, Confluence

EDUCATION

Purdue University Northwest

Master of Science in Computer Science

IN, USA

Aug 2023 – May 2025

Amrita Vishwa Vidyapeetham

Bachelor of Science in Computer Science

Coimbatore, India

Jun 2016 – Jul 2020